

Examiner reports

Q1.

The proof of compound angle formulae is new to the A-level maths specification and many students found this carefully structured question very challenging.

Part (a) was an “explain why” question and it was not enough to simply identify $\cos A$ and $\sin B$ when a key part of the proof is to show that $\angle EFQ = A$. Some students achieved one mark with a partial explanation, but it was rare to see full marks.

Around half of the students completed part (b) successfully.

In part (c) only a minority of students could identify the limitations of the proof.

Many students ignored the direction given in part (d) and set about trying to prove the identity from scratch, with no success. Most students who began the argument correctly, replacing B with $-B$ as suggested, scored full marks.

Q2.

Many students made good progress with this question and chose an appropriate technique to be able to determine the transformations. Most identified $R \sin(x - \alpha)$ as the best approach, although some students chose $R \sin(x + \alpha)$ and a few individuals started with $R \sin(x \pm \alpha)$, which often resulted in sign errors.

Having decided on an approach, most students handled well the routine of finding R and α although some used some inefficient methods and did not seem sufficiently well practiced in what is a very standard technique.

In order to fully justify their answer a clear comparison of the given equation and their $R \sin(x \pm \alpha) + 4$ needed to be made and in a few cases students' reasoning was not clear. Most students achieved some credit for correctly describing transformations relating to their $R \sin(x \pm \alpha) + 4$. All was not lost if mistakes were made early in the solution.

Part (b)(i) hinged on students realising that the denominator had to be maximised and so their $\sin(x \pm \alpha)$ had to equal 1. Students who had gone wrong in part (a) could still achieve some credit in part (b)(i). Many students gave several lines of working to rationalise the denominator of their fraction. This was unnecessary given allowed calculator technology.

The mark for part (b)(ii) was available to students who had gone wrong in part (a). Provided they used their R correctly, work was followed through.

Q3.

(a) was very well answered with around two-thirds of students scoring full marks.

(b) proved to be quite a challenging question with a wide spread of marks. A number of students failed to see the connection to (a) despite the “Hence...” in the question, and so never arrived at an appropriate integrand to integrate. There were a wide variety of methods used to integrate including inspection and various substitutions which had the

potential to be successful, and some very imaginative approaches were seen. Some students used a substitution of $x = 2\theta$ initially but failed to correctly replace $d\theta$, meaning that, despite correct later work, they were unable to arrive at a valid, correct solution.

Q6.

The majority of students did well on part (a) of this question and well over half of the cohort scored 6 marks or more. Most students were less successful on part (b), although many started by writing down several relevant looking identities.

Part (a)(i). The vast majority of students answered correctly, with relatively few finding $\alpha = 56.3^\circ$ rather than the expected $\alpha = 33.7^\circ$.

Part(a)(ii). Although most students scored the 3 marks some failed to state the minimum value where an explicit statement was required. Similarly, some students just wrote down the value of x at which the minimum occurred without justifying it. Less common errors

were to use $\cos^{-1}\left(-\frac{1}{R}\right)$ or to assume the minimum was at $x = 90^\circ$.

Part (b)(i). Many students wrote the given expression in terms of $\sin x$ and $\cos x$, but then went on to treat the identity as if it were an equation, often equating to zero. Those who used a common denominator or took out a factor of $\frac{1}{\sin x}$ usually completed successfully.

Part (b)(ii). Although most students recognised that they had to use the result from part (i), many failed to consider both $\cot x = 0$ and $\cos 2x = 0$ and so gained no credit. Of those who did consider both these equations many students thought $\cot x = 0$ had no solutions, or they changed it to $\tan x = 0$. Most students did solve $\cos 2x = 0$ correctly, although not always giving both solutions within the given range.

Q7.

The response to this question was very mixed, due to a student's understanding, or misunderstanding, of the term exact. Despite $\tan \alpha$ being given as a surd in the question, many students interpreted *exact* as meaning a number with many decimal places. As a result, many students, nearly 30% of the entry, scored at most 2 of the 8 marks available, only gaining credit for showing some knowledge of the required trigonometric formulae and nothing else. A similar proportion, however, had the question fully correct.

- (a) (i) Angle α was given as acute and it was expected students would use a right angle triangle to show the requested value of $\sin \alpha$. Many did this, but many found the angle itself using inverse tangent and then its sine for no credit;

$$(\tan^{-1} \frac{2}{\sqrt{5}}) = \frac{2}{3}$$
 similarly $\sin$

gained no credit. Many found $\cos \alpha$ from $\sin \alpha = \cos \alpha \tan \alpha$ which was accepted, if exact. Some students successfully derived the result using $\operatorname{cosec}^2 \alpha = 1 + \cot^2 \alpha$.

- (ii) The vast majority of students knew, or could derive, the double angle formula for $\sin 2\alpha$, and so obtained the correct exact value if they had $\cos \alpha$ correct.
- (b) Most students knew the required compound angle formula, although occasionally there was a sign error and some thought the formula was $\cos \alpha - \cos \beta$. Many

approaches were used to find $\sin \beta$ and $\cos \beta$. Some used the simplest method,

involving a right angle triangle, whilst some noticed that $\sin \beta / \cos \beta = \frac{1}{2}$ and used this in conjunction with $\sin^2 \beta + \cos^2 \beta = 1$ to find the values. Most students who had the four component parts correct could complete to the requested form; but some

were unable to manipulate the surds as required, even from $\frac{2}{3} + \frac{2}{15}\sqrt{5}$. Many students who had worked with inexact values, wrote down $k = 5$ as their final answer, but such students need to understand that using results from their calculator does not show this, as requested in the question.

Q8.

This question was generally done well with about 35% of the entry gaining 10 or 11 marks. Often the mark lost was in part (a) for failure to show any processing or to give a conclusion.

- (a) Virtually all students knew they were to substitute $x = -\frac{3}{2}$, and evaluate. However, those who just substituted into $f(x)$ and wrote $= 0$ without showing any stages in the arithmetic did not score the accuracy mark, nor did those who did not interpret the $= 0$ result.
- (b) Most students had this correct and virtually all scored at least the method mark. Some students did the factorisation by inspection whilst long division was also a popular method. Those students who chose to use undetermined coefficients often did much unnecessary work, as at least the values of a and c should be apparent from inspection.
- (c) (i) This part of the question was also generally done well with most students able to quote the cosine double angle formula or derive it correctly, and convert it to sines.
 A rare error was a sign error or the dropping of the 2. Some students did not handle multiplying out the expression convincingly, and as the required result was given, they were penalised. However, most did convert their trigonometric expression into the required equation in x .
- (ii) Some students did not attempt to solve this equation, but of those who did they virtually all knew they had to solve the cubic equation in x . Many did this successfully, often following a failed attempt at factorisation. Some made sign or other errors in their use of the quadratic formula. Some students used the quadratic formula on the cubic equation itself. Students who had not used the quadratic formula correctly with the given equation, could gain no further marks. However, many did complete successfully and it was pleasing to see the inappropriate roots of the cubic equation carefully rejected. Some students did not correctly convert the -16.3° they obtained from their calculator into angles in the required range.

Q9.

Most candidates had part (a)(i) correct, although the amount of work they did varied considerably. Some just wrote down the value of $\tan \alpha$, presumably recognising the 3-4-5 triangle, whereas others used Pythagoras. Several weaker candidates divided $\sin \beta$ by

$\cos \alpha$ to give an answer of $\frac{5}{6}$. Some candidates used trigonometric identities, some finding $\sin \alpha$ first and using it together with the given cosine to calculate the tangent.

In part (a)(ii), the majority of candidates failed to work with a negative value for $\tan \beta$, despite most of them showing awareness of an obtuse angle through quoting 150° . These candidates then could not get to the right answer in part (b).

However, in part (b), most candidates did use the compound angle formula for $\tan(\alpha + \beta)$ correctly, with only a few sign errors seen and some leaving the $\pm$, from the formula book, in their expression. Most tried to manipulate the surds in their expression towards the required form of the answer, but with varying degrees of success. Those who inverted the denominator and then multiplied the resulting fractions were the more successful. Some who had worked with the correct value of $\tan \beta$ failed to get to the required form by not seeing that they could cancel or multiply through their expression by a factor of $\sqrt{3}$.

Q10.

In part (a), virtually all candidates knew that they were to substitute $x = \frac{3}{4}$ into the cubic expression, but many either failed to show any detail of a calculation leading to zero and / or failed to give a conclusion as to the significance of such a result. Candidates need to recognise that when a printed answer is given, as with $(4x - 3)$ being a factor here, they need to give a full justification of why this is the case.

In part (b), most candidates showed that they were familiar with the double angle formulae, but more so the double sine formula, with there often being an error in that of the double cosine. However, many substituted correctly, and manipulated their expression in θ correctly using $\sin^2 \theta = 1 - \cos^2 \theta$, before converting to x . Some substituted x for $\cos \theta$ as soon as they could and sometimes got into a confused expression which they were unable to deal with convincingly. Some candidates tried to approach the problem by working on the sine and cosine parts separately; sometimes they were successful but again some became confused when they tried to put it back together in terms of x . Some candidates failed to secure the final mark by not equating to zero.

Most candidates attempted part (c) and knew that they were to factorise the cubic expression. Some used long division by the given factor of $(4x - 3)$ but then did nothing with their resulting quadratic expression, whereas others showed the correct interpretation and wrote down the factorised form of the cubic expression. Some candidates factorised the cubic expression by inspection, mostly correctly, but with the occasional error in the x -term of the quadratic factor. Some candidates who had a correct factorisation failed to score the last two marks as they simply said that the quadratic did not factorise or that it had no roots, without demonstrating why. To score the marks, candidates were required to calculate their discriminant and, if correct, note that it was negative and thus that there were no more (real) solutions to the cubic equation. Most of those who did this concluded the question with a convincing argument as to why there was only one solution to the cubic equation and thus to the trigonometric equation and stated the real solution.

Q11.

In part (a), whilst most students found the value of R correctly and left it as $\sqrt{10}$, the same was not true of the angle α . Some students seemingly failed to notice that α was given as an acute angle and found $\tan \alpha$ to be -3 ; some confused sine and cosine and had $\tan \alpha = \pm \frac{1}{3}$. Some students progressed to $\tan \alpha = 3$ from $\sin \alpha = 3$, $\cos \alpha = 1$ which was penalised for both marks.

For part (b), most students rearranged the equation correctly using their result from part (a), although many then dropped the negative sign when finding the inverse sine. Many students seemingly could not cope with the result of -39.23° that their calculator gave them, when all that was required was to add 71.6° (ie α) to get the first solution. Some students found the third quadrant solution only, whereas some found more than two solutions. There were many students who ignored the request to round to the nearest degree in their final answer; this was condoned if their answers were correct to the greater number of decimal places they gave. Many students did give a fully correct solution, often by using a sketch graph to help them decide where the solutions lay.

Q12.

This question provided a good start for the candidates with over half the entry gaining full marks, although some candidates omitted it altogether.

In part (a) the most common error was to equate $\tan \alpha$ to $\frac{2}{5}$ instead of $\frac{5}{2}$. Some candidates didn't make explicit their value for R so didn't answer the question, and so lost a mark even if they gave the correct answer to part (b)(i).

In (b) despite having quoted R as $\sqrt{29}$ in part (a), some candidates went on to state the maximum of $2\sin x + 5\cos x$ as 5, and said the maximum occurred at $x = 0$, or other similar errors. Some candidates had the correct answer for where the maximum occurred in their answer, but then went on to give more than one value, and so lost the mark. A few candidates correctly found the minimum value and where it occurred, but gained no credit.

Almost all candidates heeded the instruction to give values of angles to the nearest 0.1° .

Q13.

There was a wide range of responses to this question and relatively few candidates gained full marks.

In part (a)(i) most candidates wrote down a correct form for $\tan 2x$ with only a few writing $\tan^2 x$ instead of $2\tan x$. Most multiplied by the denominator to obtain a cubic equation in $\tan x$. However, many divided throughout by $\tan x$ to get $\tan^2 x = 3$ and failed to demonstrate that $\tan x = 0$ is also a solution. Some used verification which did not warrant any marks.

In (a)(ii) very few candidates gained this mark. The solution of 120° was almost always omitted. Most wrote 0° and 180° too. This in itself was not penalised this time as these values were outside the range of x requested.

Part (b)(i) was answered very well, as candidates showed familiarity with double angle formulae and handled the algebra involved with confidence.

In part (b)(ii) the quadratic formula was used correctly by most candidates and most demonstrated where the $\sqrt{3}$ came from. The majority of candidates, though, just seemed to ignore the negative root and gave no explanation as to why. If they found only one root then they lost 2 marks. Some gave unsatisfactory explanations for the exclusion of the negative root, such as $\sin x$ cannot be negative or that it was > 1 . Those who used completion of the square to solve the equation almost invariably just took the positive square root.

Q14.

Most candidates scored the marks for the two derivatives with relatively few omitting the minus signs. Some candidates managed to drop the 2 from $\cos 2\theta$. Many candidates brought the

commonly used parameter t into their expression using $\frac{dx}{dt} = -\sin \theta$ or similar, but were not penalised. Virtually all knew how to use the chain rule, but many got mixed up with the coefficients in both expanding $\sin 2\theta$ and simplifying the expression. The common error was to find $k = 3$. Some candidates got a result in terms of $\sin \theta$ and some got no result at all, but any value of k , however found or invented, was allowed in part (a)(ii).

Part (a)(ii) was generally done well with most candidates demonstrating they both knew and could use the relation between the gradients of the tan gent and the normal, although a few

found an equation of the tan gent. The evaluation using $\theta = \frac{\pi}{3}$ was generally accurately done; however candidates had to have derived $k = 6$ correctly to gain the final mark for the equation of the normal.

In part (b) it was expected here that candidates would note the whole question was about the use of double angle formulae, and many did. Most got as far as at least trying to express $\sin^2 x$ in term of $\cos 2x$ and of those, many did it correctly, even if sometimes there was a mix of variables. The integration of $\cos 2x$ was usually accurate, with the common error being to double $\sin 2x$ rather than halve it. There were few sign errors. Virtually all candidates who achieved an

integral knew how to use the limits, but the real test here was in handling $\sin 2\left(-\frac{\pi}{4}\right)$; many candidates lost credit through not showing their evaluation or getting the sign wrong.

However, many candidates, about 25% of the entry, went on to obtain the correct answer and give it in an exact form, although some did make a mistake with the signs, particularly the three minus signs involved with the lower limit. Of the few candidates who attempted the integral using parts most made little progress beyond the first line of working, not realising that in order to proceed with the integral they needed to change $\cos^2 x$ back to $\sin^2 x$, and manipulate the resulting expression.

Q15.

In part (a), most candidates were able to find R and α correctly, with only a few arithmetic errors seen. Some candidates made the error of using $\tan \alpha = \frac{1}{3}$, or equivalent, although most candidates did use the tangent to find the value of α . Some ignored the request for three decimal places, but two decimal place accuracy was condoned.

Many candidates obtained a correct answer to part (b), although some did not give the minus sign on $\sqrt{10}$, or thought 0 or -1 was the minimum value. There was confusion among some of the candidates between the minimum value and the value of x at which it occurred, although again many correct answers were seen, but here too not always to three decimal place accuracy.

In part (c), candidates who realised that they were to use part (a) usually made some progress. Those relatively few candidates who did not realise this usually tried to square the equation, but many did not realise this could only be successful if the sin or cos term was moved to the other side first. Many such attempts were abandoned. Those who used the expected method often lost 1 or 2 marks, through not giving all possible solutions in the range, and in particular the last accuracy mark was often lost as candidates did not make use of -0.886 in their working. Three decimal place accuracy was required for the final mark as were answers in radians: degrees were condoned during the working and in part (b) but not in part (a).

Q16.

In part (a)(i), most candidates stated the formula for $\sin 2\theta$ correctly although not always in its simplest form.

In part (a)(ii), although many candidates knew the formula for $\cos 2\theta$ in at least one its correct forms, several gave $\cos^2 \theta + \sin^2 \theta$, and later in the question just gave the value of $\cos 2\theta$ as 1 without apparently thinking something could be wrong.

In part (a)(ii), many candidates had problems. It appeared that the 3-4-5 triangle was either not well known or not recognised as relevant; although some candidates did draw this to justify their answer others used $\sin^2 \theta = 1 - \cos^2 \theta$. Many candidates worked backwards from the answer given, to find a value for $\sin \theta$, which gained no credit nor did going via an angle from the given value of $\cos \theta$. Many candidates omitted to find $\cos 2\theta$ or, as above, gave the value as 1, or another incorrect value from an incorrect expression. Those who did find the correct value for $\cos 2\theta$ often dropped the minus sign in later working.

In part (b)(i), the use of the chain rule with these parametric equations was generally done well with most candidates completing it correctly. Some candidates apparently tried to integrate rather than differentiate and some made sign errors when differentiating. A few candidates had the chain rule expression the wrong way up.

In part (b)(ii), although some neat and concise correct answers were seen, many candidates got in a mess trying to use angles and not working in exact fractions. Some confused the value of the sine and cosines involved with the angles writing terms such as $\sin \frac{3}{5}$. Some left a mix of fractions and decimal numbers in their final answer, which is bad practice but was accepted if the values of the coordinates at P were correct.

Q17.

- (a) (i) Most candidates either knew the appropriate identity for $\cos 2x$ to use, or worked it out in terms of $\sin x$ from $\cos^2 x - \sin^2 x$. The common error was to attempt to use an “identity” $\cos 2x = \sin^2 x - 1$ or similar, and candidates did gain credit for attempting to get an expression in $\sin x$. Most candidates, who had used a correct identity, went on to convincingly obtain the given equation. Others, who had an incorrect identity, chose to ignore inconsistencies and just wrote down the given result. Those few candidates who had “invented” an identity that led to the result by working backwards from the answer given, gained no credit.
- (ii) Most candidates used factorisation correctly to get the required results. Those who attempted to use the quadratic formula were more prone to making an error. Some candidates attempted to solve their equation from part (a)(i) rather than the given equation, and gained no credit. Many candidates wasted some time here by proceeding to find the angles, which had not been asked for.
- (b) (i) The technique for finding R and α was well known and most candidates did this correctly. The common error was to find α as 56° , and less so a numerical error in finding R . Some candidates appeared not to notice they were given $R > 0$, and gave $\pm R$ in their result. Another error sometimes seen was for candidates to confuse $2x$ and α , and thus get half the expected value for α .
- (ii) Although most candidates knew they were to use their result from part (b)(i) to solve this equation, many made an error in manipulating the given equation to a soluble form. Either 0 or 1 appeared on the right hand side rather than -1 and such candidates could gain no further credit. Whilst many candidates proceeded to solve the equation for one correct angle, relatively few got both correct and some got neither correct. Those who drew a sketch of the cosine curve, usually found it helpful in realising there were two solutions. Those few candidates who did not attempt to use the result from part (b)(i) to solve the equation gained no credit. Attempts just by using a calculator were very rare, and there had to be evidence of using part (b)(i) to gain any credit.

Q18.

Some candidates clearly had little knowledge of this part of the specification, whilst others gave clear and concise solutions. Part (a) was more successful than part (b), although many made a sign error in comparing their expansion with the given expression and found

$$\tan \alpha = -3 \text{ or } \tan \alpha = \frac{1}{3}.$$

Very few errors were seen in finding the value of R , although many did not realise that $-R$

was the minimum value. This was frequently given as 0, -1 or $\frac{\pm 1}{\sqrt{10}}$, and it was often confused with finding the value of x where the minimum occurred.

Some candidates made the mistake of using $-\frac{\pi}{2}$ instead of $\frac{3\pi}{2}$ and gave a result which was out of range. Most candidates did work in radians to the requested accuracy, and

some drew sketch graphs to aid their solution.

Q19.

In part (a), nearly all candidates quoted the $\sin 2x$ expansion correctly, and attempted to use it in part (b). In part (b), many cancelled the $\cos x$ terms without considering $\cos x = 0$, and so lost a mark. Some candidates, in dividing by $\cos x$, brought $\tan x$ into the equation

and scored no further marks. Most candidates went on to find $\sin^{-1}\left(-\frac{1}{3}\right)$, although a few made a sign error. The calculator solution of -17.5° was interpreted correctly by most candidates, to give solutions in the required range. The responses to the 'proof in part (b)' varied considerably. Some candidates seemed to see immediately that they needed the $1 - 2\sin^2 x$ version of $\cos 2x$ and obtained the requested result very concisely, and it was nice to see them justifying division by $\sin x$.

Others, although not making any mistakes, took rather longer to arrive at the result, some replacing 1 with $\sin^2 x + \cos^2 x$. Some candidates made no progress beyond quoting $\cos 2x = \cos^2 x - \sin^2 x$. For those who did make further progress, failure to complete was usually due to a dropped 2 or 1, or just losing sight of what they were trying to do. Those few candidates who worked with just the left hand side of the expression usually could not establish the result.

Q20.

There were many correct answers to part (a)(i). Rather than using $\cos 2x = 2\cos^2 x - 1$ directly, some answers were obtained from $\cos 2x = \cos^2 x - \sin^2 x = \cos^2 x - (1 - \cos^2 x)$. This sometimes led to sign errors. Other errors were the use of $\cos^2 x - 1$ or failure to multiply the -1 by 3.

In part (a)(ii) most candidates factorised successfully but then wasted time solving for x . The question did not ask for solutions. Those who used the quadratic formula often made mistakes.

Candidates generally demonstrated that they had both the knowledge and appropriate skill to answer part (b)(i) successfully and there was a pleasing number of correct solutions. The most common error was to write $\tan \alpha = 7/3$ instead of $3/7$.

In part (b)(ii) not all candidates read the request to give answers correct to the nearest 0.1° . These candidates were only penalised once. A common incorrect answer was 171.5 from $180 - 8.5$ instead of $180 - 31.7 - 23.2$. Clearly there was confusion when dealing with the obtuse angle. Some candidates changed $\sin(\theta + \alpha)$ to $\sin(\theta - \alpha)$, so could not complete successfully.

In part (c)(i) most candidates did not realise the need to use exact values and they used a calculator to find angle β and then its cosine. This gained no credit. Those who drew a diagram fared better as they often went on to use Pythagoras' theorem in an appropriate form, which showed where the value of $1/3$ came from.

Marks could only be earned in part (c)(ii) if $\sin 2\beta = 2\sin\beta\cos\beta$ was seen. Some candidates used calculators once more and did not give an exact answer. Some did realise that 0.44444 was $4/9$ but didn't gain any credit unless the double angle sine formula had been

used. Many candidates made the mistake of apparently thinking $\sin 2\beta$ was $2\sin\beta$.

Q21.

There was a full range of marks seen in response to this question. Some candidates were able to complete the whole question successfully, including finding all four solutions to the particularly demanding part (a)(ii). Other candidates just scored 1 or 2 marks, from part (a)(i) and/or a correct identity in part (b)(i). Most candidates knew what to do for part (a)(i)

and got the value of R correct, although a few got $\sqrt{80}$. Many, however, had $\tan \alpha = \frac{6}{8}$. Most also knew that they were to use the expression from part (a)(i) in part (a)(ii), although a few attempted to manipulate the given expression with double angle formulae and usually abandoned the attempt. Of those using the expected approach, most scored 2 marks, some getting a third solution as well but not many finding all four. An error which was sometimes seen was to replace $2x + 53.1$ with $\theta + 106.2$.

Part (b) was generally more successful, with most candidates scoring a mark for at least one of the identities for $\sin 2x$ or $\cos 2x$ correct and many going on to use these convincingly in obtaining the requested result. Others made errors in signs and coefficients and fudged the result rather than looking for their errors. Some candidates, however, did seek an error and attempted to overwrite their work rather than starting again, often making it difficult to read what they had actually intended as their answer.

Most candidates who attempted part (b)(ii) scored at least 2 marks — the starting equation and the solution $x = 45$ — but many also obtained all four solutions. Those candidates who ignored the *hence* and again expanded the left-hand side were largely unable to complete the question successfully, although some did succeed via this route.

Q22.

It was notable in this question that candidates tended to be successful in one part or the other but not both, although some good fully correct solutions were seen.

In part (a) many clear derivations of the given result were seen. These often involved some unnecessary steps, such as returning to double angles to eliminate $\cos^2 x$. Those who didn't complete this part successfully usually had an error in the double angle formulae, often omitting a 2. Many candidates wrote down the given answer even though this clearly did not follow from the previous line.

For part (b) most candidates knew they were expected to use the result from part (a) to do this integral; although some just ignored it, with $\sin^4 x$ often appearing in such solutions. Few correct alternative methods to the one expected were seen. The common errors in the expected method were in sign and coefficient errors in the integrals of $\sin x$ and $\sin 3x$.

Q23.

It was anticipated that candidates would find most of this question accessible, but many made mistakes or showed misunderstandings of what the question parts required.

In part (a)(i) relatively few candidates could just write down the result for $\cos \alpha$; some sensibly drew a right angled triangle to help them, whilst others derived the result from $\cos^2 \alpha + \sin^2 \alpha = 1$. Many candidates, though, found the angle on their calculator and then

its sine, some giving the result as 0.600.

Most candidates gave a correct expansion of $\cos(\alpha - \beta)$ for part (a)(ii) but then left that as their answer, without substituting the now known values of $\sin \alpha$ and $\cos \alpha$.

Relatively few candidates responded to the word “exact” in part (a)(iii) by using the compound angle formulae to give $\cos(\alpha - \beta)$ as a rational number. Some candidates confused the angles with the values of their sines and cosines. Other candidates just calculated angles, and gave an interpretation of “exact”, as the requested cosine given to many decimal places.

In part (b)(i) most candidates used the double angle formula for tangent successfully, and went on to derive the given quadratic equation. The rarely seen errors were to confuse the denominator and numerator in the expression for $\tan 2x$ or to actually leave $\tan 2x$ in the expression in place of $\tan x$. Having derived a quadratic equation, few candidates proceeded to try to solve it by conventional methods in part (b)(ii) and some did not attempt this part of the question at all. Many manipulated the equation, either in $\tan x$ or

$\tan 22\frac{1}{2}^\circ$, in various ways before abandoning it. Some made progress through completing the square but often forgot to consider both the positive and negative square root. Of those who did solve the quadratic equation correctly, few could give a convincing justification for rejecting the negative root, usually just writing something to the effect that the result must be positive, if they wrote anything at all.

Q24.

Most candidates just wrote down the required expression for $\cos 2x$, although some worked it out from $\cos^2 x - \sin^2 x$ or $\cos(A + B)$. Omission of the 2 was a relatively rare error. Most candidates were able to derive the result in part (b)(i) although some were penalised for being too brief. Most candidates too were able to solve the resulting quadratic equation, with some making it a little more difficult than intended by reading the 1 on the right hand side as 0. Most candidates found two results for angle x in the required range. Part (c) was less successful: although most candidates understood that they needed the integral in terms of $\cos 2x$, there were many algebraic slips, and sign and coefficient errors in the integral of $\cos 2x$ itself.

Q25.

This question proved to be more straightforward than anticipated, with many candidates gaining full marks from a well-presented answer. Most candidates could do part (a), although a few did not simplify their answer, although they were not required to. However $\tan 2x$ expressed as $\tan(2 + x)$ was seen a few times. In part (b) most candidates took the anticipated route of substituting for $\tan 2x$, and then convincingly dealt with the resulting algebraic fraction. Some then made a sign error in their simplification. Some manipulated the expression for $\tan 2x$ into an expression for $\tan^2 x$ and then substituted that into the expanded form of $(1 - \tan x)^2$. Some did this correctly, but others got bogged down due to algebraic errors and often abandoned their attempt.

Q26.

Part (a) was done well by most candidates, although some did a lot of unnecessary work

in obtaining the correct answer. It should be possible to write down the values of R and $\tan \alpha$ from the expression.

For part (b), most candidates understood what was expected, with many getting both solutions correct. Others were unsure whether or not there was a second solution, and often did not give one or gave more than one. A common wrong answer was 256.7° , from $360^\circ - 103.3^\circ$.

Although some candidates answered part (c) confidently, many did not and some did not attempt it. There was confusion between maximum and minimum, and the common errors were to say that the minimum occurred at 90° or 180° with the expression equated to 0 or -1 and not -5 . Some thought the minimum value was an angle. The few attempts to use calculus were usually unsuccessful with candidates again unable to demonstrate that they had found a minimum value.

Q27.

Parts (a) and (b) were done well and there were few errors, although some did make a sign error in finding angle α but didn't respond to the given answer, and just made a comment about it being an acute angle 'so is positive'.

Part (c) was found difficult by many candidates. Although most realised they were to use the results of part (a) rather than calculus, they confused the cosine having a maximum of 1 with the maximum value of the expression. Some candidates who had shown the maximum occurs at -33.7° , then subtracted from 90° or 180° to get a positive angle.

Q28.

Part (a) was usually correct. Some candidates just wrote the result down whilst others derived it from $\cos^2 x - \sin^2 x$. The common error was to have $2 \cos^2 x + 1$, or the 2 missing. There were many fully correct responses to part (b) and nearly all candidates made some attempt to make use of the double angle in their integration. Accuracy sometimes slipped with the $1/2$ attaching to the $\cos 2x$ term only. Virtually all candidates gave evidence of correct use of limits, but those who didn't obtain the requested form of $\frac{\pi}{\alpha}$, showed no attempt to find their error.

Q29.

This question in general was not done well. A lot of candidates indicated a poor knowledge of trigonometric identities and a lack of experience in manipulating them. The Specification states that the double angle formulae should be learnt.

Part (a) Although most candidates could recollect or work out the identity for $\sin 2x$ this was not the case for $\cos 2x$ where many variations on incorrect answers were seen; some were correct but not in the requested form.

Part (b) Candidates were given credit for attempting to use their versions of $\sin 2x$ and

$\cos 2x$ together with $\tan x = \frac{\sin x}{\cos x}$. Those who started on the left hand side often omitted a common denominator of $\cos x$, some just putting it back into their work to get to the given answer. Those who started on the right hand side were generally more

successful, but most candidates worked from both sides together, often making errors in their manipulations, such as omitting brackets and thus not multiplying out correctly. Many managed to make the two sides become equal after some invalid manipulation.

Part (c) Most candidates made the solution of this equation far more involved than was intended or is necessary. Few realised they could use the answer from part (b) and make use of, if $pq = 0$ then $p = 0$ or $q = 0$ and thus write down the solutions, with little more work. Some candidates did write down some correct solutions with no working seen, which was accepted. Many manipulated either form of the equation, into sines and cosines often making errors and abandoning the attempt. Many derived the cubic equation $\sin x - 2\sin^3 x = 0$ or something close to it, but still did not realise that $\sin x = 0$ was a solution. Of those who did get to correct solutions, the solution 225° , was often missing.

Q30.

There was a very mixed response to this question with relatively few candidates gaining full marks. Some didn't know how to start it at all.

Part (a) Most candidates found R successfully but made errors in finding the value of α with some finding α to be 63.4° . Most candidates responded to the request for α to be to one decimal place, though this created problems for some in part (b).

Part (b) Many candidates gave answers close to, but not exactly, zero, not realising they were rounding off the same angle to different accuracies. Some candidates did not

progress beyond $\sin(x + \alpha) = \frac{\sqrt{1}}{\sqrt{5}}$ and, of those who did, many had problems in finding the second solution.

Q31.

Part (a) Most candidates successfully wrote down the expression for $\sin 2x$.

Part (b)(i) This caused difficulties for some candidates. Fairly common answers were $2\cos 2x = 2\cos^2 x - 2\sin^2 x$ or $\cos 2x = \cos^2 x + \sin x$.

Part (b)(ii) Most candidates made a reasonable start at proving the identity with many more able to handle the given $x + 2x$ than could consider $x + x$ for themselves. Candidates' responses varied considerably from clear concise derivations, to dead ends and restarts, to long meandering routes; some expanding $\cos 2x$ and then bringing it back in. Most candidates knew they had to eliminate $\sin x$ at some point, but this was not always done convincingly.

Part (c) Those candidates who saw the connection to part (b)(ii) were usually successful in their integration, although there were some sign errors and also errors in handling the 4. Some candidates wrote $\cos x^3$ as $(1 - \sin^2 x) \cos x$ and went on to integrate correctly. Attempts by parts, and even by substitution, went awry. Many candidates opted for the somewhat hopeful $\sin^4 x$, or similar, as their integral but could get some credit for correct use of the given limits.

Q32.

While there were some good proofs of the formula for the range, there were also a number of confused approaches. Several of these were based on the formula as $v^2 = u^2 + 2as$. However, those who found the time of flight first often made quite good progress. Some candidates misquoted the double angle formula. Another area of difficulty was in the solution of the quadratic equation to find the time of flight.

In part (b), almost all of the candidates gave the answer 45° , but very few were able to give a convincing argument.

Part (c) was generally done very well. Some candidates tried to do this without using the results from parts (a) and (b). Many of these candidates were unsuccessful. A few candidates did not give their answer for the range to the nearest metre as instructed in the question.



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